

Supplementary Material: Cheap Metrics First, Replanning Diagnostics Second

Hyun Joon Cho, Taehwan Yun
Sungkyunkwan University

TABLE I

PRIMARY POOL SPLIT BY POLICY FAMILY. BOTH HALVES ARE SHOWN BECAUSE THE POOLED ESTIMATE IS A WASH ONLY AFTER THEY ARE AVERAGED: PLV-1 LEADS EVERY ENDPOINT ON DIFFUSION POLICY BUT IS THE WEAKEST SMOOTHNESS VARIANT ON BC-RNN, WHERE OPEN-LOOP SMOOTHNESS LEADS; OBJECTIVE LOSS IS ADVERSE ON DIFFUSION POLICY AND COMPETITIVE ON BC-RNN. THESE ARE DESCRIPTIVE SUBGROUP ESTIMATES OVER FIVE TASK MEANS WITH NO MULTIPLICITY CORRECTION, AND EACH INHERITS THE SAME RESOLUTION LIMIT AS THE POOLED COMPARISON. HIGHER ρ IS BETTER; MMRV AND REGRET@30 ARE LOWER-BETTER.

Metric	$\rho \uparrow$	MMRV $\downarrow$	Regret@30 $\downarrow$
<i>BC-RNN (21 runs)</i>			
Action L1	0.372	0.190	0.050
Open-loop smoothness	0.401	0.184	0.028
PLV-1 smoothness	0.303	0.231	0.084
PLV-1 uncertainty+smooth	0.344	0.224	0.076
Objective validation loss	0.360	0.261	0.034
<i>Diffusion Policy (24 runs)</i>			
Action L1	0.462	0.120	0.017
Open-loop smoothness	0.418	0.149	0.010
PLV-1 smoothness	0.556	0.093	0.006
PLV-1 uncertainty+smooth	0.502	0.102	0.008
Objective validation loss	-0.243	0.316	0.083

I. FROZEN ANALYSIS CONTRACT

The analysis unit is a task–dataset–policy–seed training run. Correlations are computed among checkpoints within each run, averaged over runs within a task, and then averaged with equal task weight. Task-clustered bootstrap intervals resample tasks and their nested runs. A run is outcome-resolving when its success IQR is at least 0.10, it contains at least five distinct success values, and its best–worst success gap is at least 0.20.

For Regret@ q , the retained count is $\lceil qN \rceil$; Regret@1 instead retains exactly one checkpoint. Lower-cost metrics are sorted first; boundary ties are broken by checkpoint path in the frozen manifest, never by rollout success. Every primary checkpoint has 50 simulator rollouts, so observed success is quantized in increments of 0.02.

MMRV follows SIMPLER: for each checkpoint, we find the largest true success margin among checkpoints that the metric orders incorrectly, then average these maximum violations. Lower is better. The primary task-balanced values are 0.153 for Action L1, 0.160 for PLV-1 smoothness, 0.166 for open-loop smoothness, and 0.290 for objective validation loss.

II. POLICY-FAMILY SPLIT AND DESIGN RESOLUTION

Five task means bound how small a difference this design can resolve. At 80% power the minimum detectable paired

difference is: PLV-1 smoothness versus Action L1 (observed $\Delta\rho = 0.016$, detectable $\Delta\rho \approx 0.117$); PLV-1 smoothness versus open-loop smoothness (observed $\Delta\rho = 0.019$, detectable $\Delta\rho \approx 0.101$); open-loop smoothness versus Action L1 (observed $\Delta\rho = -0.003$, detectable $\Delta\rho \approx 0.073$). No observed gap reaches 0.19 of the difference the design could detect, so the cheap-metric comparisons are unresolved at this sample size rather than demonstrated equivalences. Increasing checkpoints per run does not help: the unit of inference is the task mean.

Curvature is also present in every metric–success relation: adding a quadratic term raises R^2 by 0.15–0.19 over a linear fit with a consistently negative quadratic coefficient, and LOWESS finds 1.4–1.7 turning points per run on average. Rank correlation is therefore the appropriate summary, and a linear read of these curves would overstate late-training gains.

III. SCORE NORMALIZATION SENSITIVITY

The normalization audit confirms that the score is transductive rather than intrinsically calibrated. Task-balanced run-level Spearman is 0.366 with target-pool median/IQR, 0.364 with a fixed seed reference, 0.321 with within-run empirical ranks, and 0.178 for the unnormalized component mean. Random half-pools preserve the full-pool order with mean $\rho = 0.961$, although the mean absolute normalized-score shift is 0.186. We therefore use normalization only within a declared candidate pool and do not interpret its magnitude across pools.

IV. FINITE-ROLLOUT RELIABILITY AND BASELINES

The primary archive stores each checkpoint’s number of successes but not the order of its 50 episode outcomes. Conditional on the observed count, we draw the first half from the exact hypergeometric distribution and assign the remaining successes to the second half. Across 1,000 repetitions, the task-balanced 25/25 split-half Spearman is 0.500 with interval $[0.472, 0.529]$. The Spearman–Brown estimate for all 50 outcomes is 0.667. This is a reliability diagnostic, not a correlation with an unobserved true success probability.

At a 30% retained budget, random selection has task-balanced mean regret 0.046 with randomization interval $[0.026, 0.064]$; selecting the last checkpoint gives 0.118, Action L1 0.033, open-loop smoothness 0.019, and PLV-1 smoothness 0.045. The exact paired task tests have only five nonzero task differences, so their smallest attainable two-sided signed-rank p -value is 0.0625.

TABLE II

EXPERIMENTAL COHORT MAP. COUNTS ARE NOT ADDITIVE: THE 504, 130, AND 47 ROWS ARE NESTED SUBSETS OF THE PRIMARY POOL. SELECTED GRIDS ARE NOT COMPLETE-GRID CONFIRMATION.

Cohort	Policy	Tasks/runs	Checkpoints	Rollouts	Grid	Evidence role
Primary retrospective	BC-RNN, DP	5/45	1,365	50	heterogeneous	main analysis
Execution archive	DP	5/24	504	50	archived	development
Four-seed controls	DP	5/24	130	50	retained	post-hoc control
Same-start branches	DP	4/9	47	50	retained	mechanism
Coffee/StackThree	BC-RNN	2/8	328	50	complete	frozen
Eight-task breadth	BC-RNN	8/32	1,312	50	complete	frozen
Action-chunk breadth	DP	8/8	168	50	complete	frozen
Unseen family	ACT	2/6	126	50	complete	transfer
Domain variants	ACT	2/6	126	50	complete	strict prospective
Unseen semantic family	ACT	2/6	126	50	complete	strict prospective
Strict sequence	ACT	17/51	561	50	complete	prospective
Candidate-only stress	ACT	3/9	40	50	selected	non-discriminative

TABLE III

EXTERNAL EVIDENCE MATRIX. “CONFIRM.” REQUIRES A FROZEN, COMPLETE, OUTCOME-RESOLVING COHORT; “DESCRIPTIVE” DENOTES SELECTED OR INCOMPLETE COVERAGE; “FAILED” MEANS A FROZEN PLV CLAIM WAS NOT CONFIRMED; AND “N/D” DENOTES NON-DISCRIMINATIVE OUTCOMES.

Cohort	Policy	Complete	Frozen	Resolving	Action/objective evidence	PLV evidence
Coffee/StackThree	BC-RNN	Yes	Yes	Yes	objective strongest	failed vs. baseline
Eight-task breadth	BC-RNN	Yes	Yes	Yes	objective strongest	failed vs. baseline
Action-chunk breadth	DP	Yes	Yes	Yes	open-loop strongest	below Action L1; gate not met
MimicGen selected	DP	No	Yes	Yes	not tested	descriptive
Retained recurrent	BC-RNN	No	Yes	Yes	not tested	descriptive
Unseen family	ACT	Yes	Yes	Yes	action signal	failed confirmation
Domain variants	ACT	Yes	Yes	Yes	not tested	failed gate
Unseen semantic family	ACT	Yes	Yes	Yes	not tested	failed gate
Strict RoboCasa	ACT	Yes	Yes	Mixed	not promoted	failed gates
Candidate-only	ACT	No	Yes	No	N/D	N/D

V. DIFFUSION POLICY EXECUTION CONTRACT

a) External blinding deviation: The strict prospective external evaluations are preregistered but not fully analyst-blind. Methods, budgets, gates, confirmatory analysis, and reporting code were frozen before target outcomes. Two exposure events are on record: a filesystem inventory printed validation-score values embedded in checkpoint filenames, and a live integrity audit exposed normalized partial sparse rewards for one task before aggregate ranking analysis. No episode-level outcomes, registered confirmatory decisions, all-budget endpoint reports, or paired bootstrap results were exposed, the selector script and its thresholds were already fixed, and the registered analysis was not changed afterward. We therefore do not claim that checkpoint filename scores or intermediate rewards remained fully embargoed.

The low-dimensional Diffusion Policy uses prediction horizon 16, observation horizon two, executable offset $s = 1$, stride $m = 8$, and absolute actions. The committed slice is $A_t[s : s + m]$. Action L1 compares the sampled executable slice with the aligned expert slice. The global metric uses one sampled evaluation; execution controls use four branch-matched diffusion seeds and take their median after averaging validation starts.

Teacher, model, and simulator branches share the start, old plan, stride, and sampling seed. The teacher history is read from the demonstration at the next replan point. The model history shifts by one observation for every predicted transition.

The simulator branch restores the demonstration simulator-state snapshot, executes the same committed absolute-action slice after conversion to the environment action convention, and appends the resulting observations.

Diffusion stochasticity is separated with four policy seeds. For a distance d between aligned plans,

$$J_{\text{sample}} = d(\pi(x; \xi_1), \pi(x; \xi_2)), \quad (1)$$

$$J_{\text{state}} = d(\pi(x; \xi), \pi(x'; \xi)), \quad (2)$$

$$J_{\text{total}} = d(\pi(x; \xi_1), \pi(x'; \xi_2)). \quad (3)$$

The first changes only sampling noise, the second uses common random numbers and changes only state, and the third matches stochastic deployment. A high sampling-only score is a warning about policy stochasticity, not evidence of closed-loop state sensitivity.

The clean mechanism contrasts are

$$D_{M,E} = d(A^M, A^E), \quad D_{S,E} = d(A^S, A^E), \quad (4)$$

$$D_{M,S} = d(A^M, A^S). \quad (5)$$

The completed 47-checkpoint archive stored only each branch’s distance to the old-plan suffix. It therefore supports the weaker excess proxy $\Delta J^M = J^M - J^E$, not a post-hoc reconstruction of the direct contrasts. The released evaluator now records the direct common-seed plan distances, but they are not reported as completed results in the manuscript.

VI. SAME-START AVAILABILITY AND DEVIATION LOG

The same-start audit contains all 47 locally available nonzero-epoch binaries in nine retained task–dataset–seed grids: 6,016 raw observations (47×32 starts $\times 4$ policy seeds), with no duplicate checkpoint/start/seed keys. Inclusion did not use success. Each run has five or six checkpoints; conditional residuals are finite in six of nine runs.

The launch configuration was reconstructed during execution from live worker commands and the fixed queue. After launch, the aggregation gate was tightened to require at least eight finite runs. Policy inference, dynamics training, simulator stepping, and raw row generation were unchanged. The final gate is therefore labeled post-hoc rather than preregistered.

Two raw fidelity/ordering gates pass, but the teacher-corrected gate fails. The task-balanced excess-success correlation is 0.068, model–simulator excess agreement is 0.096, and the residual after Action L1 and teacher replan is -0.309 over six eligible runs. Descriptive Regret@30 is 0.017 for the excess proxy and 0.031 for Action L1; only two task differences are nonzero, giving an exact two-sided Wilcoxon $p = 0.500$. These values do not justify automatic reranking.

Held-out demonstration calibration and policy-induced fidelity are sharply different. Task-balanced normalized-state MSE is 0.0017 at one step and 0.0045 averaged over five demonstration steps. In the 47-checkpoint same-start audit, one-step policy-induced MSE is 0.063 and final imagined–simulator MSE after 16 actions is 3.135. Raw model–simulator replan ordering is 0.519, but teacher-corrected excess agreement is only 0.096. The simulator-induced replan statistic itself has task-balanced $\rho = -0.212$, so learned-model error is not the only bottleneck.

The archived 504-checkpoint aggregation audit does not support the hypothesis that sparse failures were hidden by averaging. Direct mean replan reaches $\rho = 0.557$, compared with 0.448 for p90 and 0.418 for worst-20% CVaR; the corresponding recursive values are 0.551, 0.498, and 0.459.

VII. CONTROLLED STRIDE AND CAPACITY CHECK

The Phase 3 check reuses the same 47 retained checkpoints, 32 validation starts, and four common diffusion seeds for every condition. Starts are drawn from the common 16-action candidate set. Thus the valid replanning contrasts are stride four versus eight at horizon 16 and horizon eight versus 16 within stride four. The stride-eight, horizon-eight cell has one policy call and is reported only as a fidelity control.

At the deployed stride and horizon, width 512 changes oriented model-replan rho by -0.085 relative to width 256; only two of nine paired runs improve. The task-bootstrap interval is $[-0.245, 0.142]$ and the exact paired Wilcoxon $p = 0.625$ (the smallest attainable value for four nonzero task means is 0.125). The wider model therefore improves state prediction without establishing a ranking improvement under the same eight-epoch budget.

TABLE IV

CONTROLLED SENSITIVITY CHECK. VALUES ARE TASK-BALANCED OVER NINE RETAINED RUNS. REPLAN COSTS ARE NEGATED BEFORE CORRELATION, SO HIGHER ρ IS BETTER. MODEL-REPLAN RHO OR FIDELITY MSE ARE SHOWN ONLY WHEN DEFINED.

Condition	Replan ρ	Final MSE	Reading
$m=8, H=8, w=256$	–	0.408	fidelity only
$m=4, H=8, w=256$	-0.357	0.557	shorter horizon
$m=4, H=16, w=256$	-0.281	3.270	stride check
$m=8, H=16, w=256$	-0.125	3.135	baseline
$m=8, H=16, w=512$	-0.211	1.309	lower MSE, worse rank

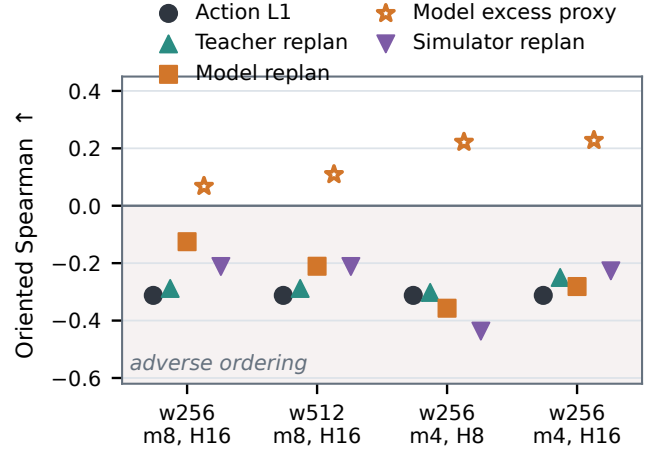


Fig. 1. Capacity and stride do not repair the model-replan result. Each point is a task-balanced estimate over the same nine same-start runs (47 checkpoints, four tasks); replan costs are negated before correlation, so higher is better and the shaded half-plane is adverse ordering. Every replan branch stays at or below zero in every condition, and doubling MLP width from 256 to 512 at the deployed stride moves the model branch further from the intended direction ($\Delta\rho = -0.085$, task-bootstrap $[-0.245, 0.142]$, exact $p = 0.625$). Hollow stars are the derived teacher-corrected excess proxy, not a measured branch score. Evidence grade: mechanism audit.

VIII. BUDGET SENSITIVITY AND OUTCOME-NOISE SCALE

Table V applies the same task-balanced aggregation and randomization protocol at a literal top-one choice and at 10% and 30% retained budgets. Random intervals are 10,000 task-balanced draws. The 30% result is not uniquely unfavorable to PLV: Action L1 is also inside the random interval, whereas open-loop smoothness is below it. PLV-1 is slightly lower than Action L1 only at top one, and is not the best metric at any listed budget.

The conditional split-half Spearman $r_{25} = 0.500$ gives the Spearman–Brown reliability $r_{50} = 2r_{25}/(1 + r_{25}) = 0.667$. Treating each metric as noiseless gives the one-sided classical attenuation approximation $\rho_{\text{true}} \approx \rho_{\text{observed}}/\sqrt{r_{50}}$. The resulting Action-L1, PLV-1, and objective-loss values are approximately 0.512, 0.532, and 0.055. This is not a rank-specific correction or a true-performance ceiling: the split is conditional on archived success counts and omits any unarchived initial-condition or environment-seed variation.

TABLE V

TASK-BALANCED REGRET BY SCREENING BUDGET. LOWER IS BETTER. BRACKETS GIVE THE RANDOM-SELECTOR 95% INTERVAL SHARED BY EACH ROW WITHIN A BUDGET.

Selector	R@1	R@10%	R@30%
Random	.207 [.158, .266]	.089 [.065, .112]	.046 [.026, .064]
Action L1	0.163	0.067	0.033
PLV-1 smoothness	0.159	0.075	0.045
Open-loop smoothness	0.150	0.063	0.019

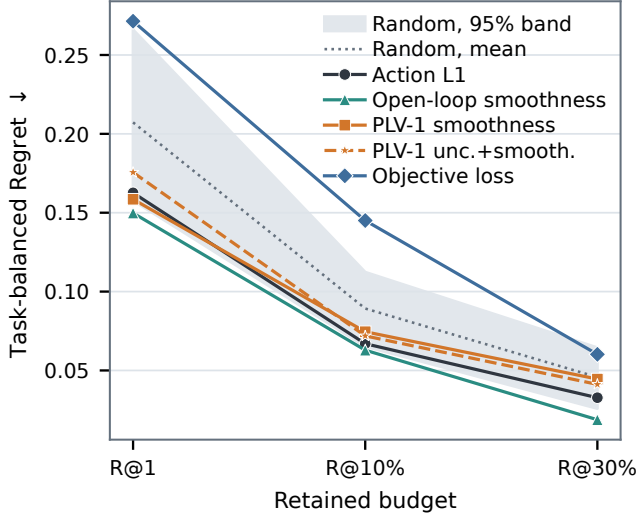


Fig. 2. Selection regret across every retained budget, so the 30% endpoint is not read alone. Lower is better. The band is the task-balanced random-selector 95% interval from 10,000 draws and the dotted line its mean; 45 runs across five tasks. Open-loop smoothness is the only metric below the random band at all three budgets, Action L1 and PLV-1 track each other closely, and objective loss sits above the band at R@1 and R@10% and inside it at R@30%. Evidence grade: primary.

IX. SCREENING RECOMMENDATION BY REGIME

The table below summarizes which signal carries each regime and what decision it licenses. It is a reading aid for the evidence already reported, not a validated deployment rule: the cascade it sketches still awaits a frozen prospective allocation trial.

a) *Strict prospective chronology*: Two cohorts register datasets, source-model hashes, environment, target tasks, and gate before target training, compute outcome-blind metrics for 126 checkpoints, and freeze those scores before collecting 6,300 binary rollout episodes. No target outcome is used to fit, select, or recalibrate the scorer. Mean Spearman moves from 0.681 to 0.629 on the domain-variant cohort (Regret@20 0.010 $\rightarrow$ 0.017, Regret@30 0.007 $\rightarrow$ 0.017, 4/6 pools stable) and from 0.304 to 0.279 on the unseen bimanual dexterous family (Regret@20 0.050 $\rightarrow$ 0.050, Regret@30 0.047 $\rightarrow$ 0.050, 5/6 stable). Both fail their complete preregistered gates. The semantic-family cohort was registered as the decisive generalization test; the domain-variant cohort is complementary evidence and does not replace it.

TABLE VI

EVIDENCE-BASED INTERPRETATION BY REGIME. THIS IS A READING AID, NOT A PROSPECTIVELY VALIDATED CASCADE; LATER STAGES DO NOT AUTOMATICALLY REPLACE EARLIER RANKINGS.

Regime	Primary signal	Decision
Heterogeneous pool	Action L1	default filtering
Matched BC-RNN	objective loss + Action L1	validate locally
DP near-tie (unvalidated)	teacher-forced replan	flag temporal disagreement
Hard/contact DP	boundary + overlap	discovery-only flag
Material disagreement	real rollout	abstain from reranking
New family / flat outcomes	recalibrate	no transfer claim

TABLE VII

TASK-BALANCED ROLLOUT SUCCESS BY WITHIN-RUN METRIC-SCORE QUINTILE (Q1 WORST SCORE, Q5 BEST). EVERY METRIC EARNS MOST OF ITS RANK CORRELATION BETWEEN Q1 AND Q3, SEPARATING BROKEN CHECKPOINTS FROM WORKING ONES, AND IS CLOSE TO FLAT ACROSS Q3–Q5 WHERE A 30% BUDGET ACTUALLY CHOOSES. EVIDENCE ROLE: PRIMARY RETROSPECTIVE. Q2 AND Q4 ARE OMITTED FOR READABILITY.

Metric	Q1	Q3	Q5	Q1 $\rightarrow$ Q3	Q3 $\rightarrow$ Q5
Open-loop smoothness	0.277	0.509	0.535	+0.232	+0.026
Action L1	0.252	0.517	0.531	+0.265	+0.014
PLV-1 smoothness	0.254	0.524	0.532	+0.270	+0.008
PLV-1 uncertainty+smooth	0.257	0.531	0.524	+0.273	-0.006
Objective validation loss	0.375	0.508	0.445	+0.133	-0.062

X. WHERE THE DISCRIMINATIVE SIGNAL SITS

Across the five metrics the climb from the worst to the middle quintile spans 0.133–0.273 in task-balanced success, whereas the entire top 60% spans -0.062 – 0.026 . Ordering the five metrics by this top-quintile resolution reproduces their Regret@30 ordering, while ordering them by global ρ does not; with only five metrics this is a descriptive ordering, not a test. The residual matters operationally: within each metric’s own best quintile the run-best success still exceeds the quintile mean by about 0.14, which is the spread no offline signal reported here resolves.

Part of the climb is training progress: within runs, metric score correlates with epoch at $\rho = 0.56$ – 0.69 for the action-space and smoothness metrics while success correlates with epoch at 0.47. Selecting the last checkpoint is nevertheless a poor rule (regret 0.118), because success plateaus and then declines late; the cheap metrics earn their value by avoiding that decline rather than by ranking the survivors. Objective validation loss is the exception in kind, correlating with epoch at -0.353 : it tracks objective overfitting, which is decoupled from deployed success and adverse on Diffusion Policy.

XI. HOW MUCH IS LEFT TO WIN

The raw spread inside the retained set overstates the opportunity. A binomial decomposition attributes a task-balanced true ability spread of 0.042 and a headroom of 0.063 above the retained-set mean, against a raw observed gap of 0.110. The 50-episode winner is the true best in only 0.322 of runs, but the checkpoints it confuses are near-equivalent, so trusting it costs 0.012. The headroom is concentrated where the tasks are hard: true spread 0.072 on Transport and Tool Hang against 0.009 on Can and Square, so on the easy tasks the retained

TABLE VIII

WHAT IS LEFT TO WIN INSIDE THE RETAINED TOP 30% (SELECTED BY ACTION L1). THE OBSERVED GAP IS THE RAW MAXIMUM MINUS THE SET MEAN; THE BINOMIAL DECOMPOSITION REMOVES THE SAMPLING NOISE THAT INFLATES A MAXIMUM OVER TEN 50-EPIISODE ESTIMATES. HEADROOM IS THE TRUE BEST MINUS THE SET MEAN, AND HIT RATE IS THE PROBABILITY THAT THE 50-EPIISODE WINNER IS THE TRUE BEST. EVIDENCE GRADE: PRIMARY.

Task	Observed gap	True SD	Headroom	Hit rate
Can	0.046	0.008	0.013	0.162
Lift	0.090	0.049	0.070	0.420
Square	0.077	0.011	0.018	0.160
Transport	0.141	0.062	0.092	0.366
Tool Hang	0.198	0.083	0.120	0.501
Task-balanced	0.110	0.042	0.063	0.322

checkpoints are close to interchangeable and no selector can gain there.

Two consequences follow. First, Regret@30 is measured against a 50-episode maximum, so part of what a selector is asked to recover is sampling noise rather than ability; the endpoint is still the right one, but its absolute scale should not be read as recoverable performance. Second, an offline selector has almost nothing to gain on Can and Square, where the retained checkpoints are close to interchangeable, and most to gain on Transport and Tool Hang, which is where the execution-matched discovery signal was strongest.

XII. EVIDENCE GRADES

a) Confirmatory support: The metric and endpoint were frozen before outcomes, the checkpoint grid is complete, the cohort is outcome-resolving, and the declared comparison is met.

b) Descriptive support: The positive result comes from selected, incomplete, availability-limited, or metric-development coverage.

c) Failed confirmation: A frozen claim or promotion gate was tested on a complete, outcome-resolving cohort and was not met.

d) Non-discriminative: The prespecified outcome-variation criterion failed, so neither positive nor negative selector evidence is assigned.

Selected MimicGen and retained recurrent results are descriptive, not complete-grid confirmations. Coffee/StackThree, the eight-task BC-RNN cohort, the action-chunk breadth cohort, and unseen ACT use complete grids. Strict RoboCasa uses frozen promotion gates but has mixed outcome resolution; candidate-only RoboCasa is non-discriminative.

XIII. ARTIFACT MAP

The source bundle’s `artifacts/` directory contains the manifest, run/pair summaries, bootstrap, raw same-start rows, Phase 1/2 Parquet tables, gates, scripts, and SHA-256 checksums. Each new Parquet table has a neighboring metadata JSON with the input hash, output hash, random seed, and Git commit. We omit the earlier cost-regret plot because runtimes lack a common hardware, batching, and reset benchmark.